

using namespace std;

```
bool contains3or7(long long x) {  
    while (x > 0) {  
        int d = x % 10;  
        if (d == 3 || d == 7)  
            return true;  
        x /= 10;  
    }  
    return false;  
}
```

Q1

```
int main()  
{  
    long long m, n;  
    cin >> m >> n;  
  
    long long count = 0;  
    for (long long i = m; i <= n; i++) {  
        if (contains3or7(i))  
            count++;  
    }  
  
    cout << count << endl;  
    return 0;  
}
```

PROBLEMS 7

OUTPUT

DEBUG CONSOLE

TERMINAL

PORT

problem1 C:\Users\bitah\OneDrive\Desktop

7

⊗ expected a '{' C/C++(130) [Ln 14, Col 5]

⊗ identifier "i" is undefined C/C++(20) [Ln 19, Col 26]

C:\Users> bitah > OneDrive > Desktop > problem2 > main()

```

1  #include <iostream>
2  using namespace std;
3  bool isNight(int hour) {
4      return (hour >= 23 || hour < 7);
5  }
6
7  int main() {
8      int d, h, m, n;
9      cin >> d >> h >> m >> n;
10
11     int s[1005], t[1005];
12     for (int i = 0; i < n; i++) {
13         cin >> s[i] >> t[i];
14     }
15
16     int totalFare = 0;
17     int prevDist = 0;
18
19     for (int i = 0; i <= n; i++) {
20         int dist;
21         if (i == n) dist = d - prevDist; // last stretch to B
22         else dist = s[i] - prevDist;
23
24         for (int j = 0; j < dist; j++) {
25             if (isNight(h)) totalFare += 5;
26             else totalFare += 3;
27
28             m++;
29             if (m == 60) {
30                 m = 0;
31                 h = (h + 1) % 24;
32             }
33         }
34
35         if (i == n) break; // done after last stretch
36
37         // Waiting part
38         for (int j = 0; j < t[i]; j++) {
39             if (isNight(h)) totalFare += 4;
40             else totalFare += 2;
41
42             m++;
43             if (m == 60) {
44                 m = 0;

```

Q2.1

I

C:\Users> bitah > OneDrive > Desktop > problem2 > main()

Q2.2

```
7  int main() {
15
16  int totalFare = 0;
17  int prevDist = 0;
18
19  for (int i = 0; i <= n; i++) {
20      int dist;
21      if (i == n) dist = d - prevDist; // last stretch to B
22      else dist = s[i] - prevDist;
23
24      for (int j = 0; j < dist; j++) {
25          if (isNight(h)) totalFare += 5;
26          else totalFare += 3;
27
28          m++;
29          if (m == 60) {
30              m = 0;
31              h = (h + 1) % 24;
32          }
33      }
34
35      if (i == n) break; // done after last stretch
36
37      // Waiting part
38      for (int j = 0; j < t[i]; j++) {
39          if (isNight(h)) totalFare += 4;
40          else totalFare += 2;
41
42          m++;
43          if (m == 60) {
44              m = 0;
45              h = (h + 1) % 24;
46          }
47      }
48
49      prevDist = s[i];
50  }
51
52  cout << totalFare;
53  return 0;
54 }
55
```


> Users > bitah > OneDrive > Desktop > problem3 > main()

```

1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  int main() {
6      string s;
7      cin >> s;
8      int n = s.size();
9      long long total = 0;
10
11     for (int mask = 0; mask < (1 << (n - 1)); mask++) {
12         long long sum = 0;
13         long long curr = s[0] - '0';
14
15         for (int i = 0; i < n - 1; i++) {
16             if (mask & (1 << i)) {
17                 sum += curr;
18                 curr = s[i + 1] - '0';
19             } else {
20                 curr = curr * 10 + (s[i + 1] - '0');
21             }
22         }
23         sum += curr;
24         total += sum;
25     }
26
27     cout << total << endl;
28     return 0;
29 }

```

Q3